

OPIM639 · Innovations & Operations of Intelligent Online Marketplaces

Chagee: opening overseas stores right

Predict demand, provision the store, size the counter

Group [TBC]

[Member names TBC - confirm with the team]



Chagee's supply chain is fixed, so new-market risk lands on how much a store opens with and how fast its counter serves



Scaling fast into markets it has not run before

7,453 stores at end-2025, 345 of them overseas in 7 markets, and about 200 more overseas openings planned for 2026. FY2025 GMV RMB 31.58bn, net revenue RMB 12.91bn.



The upstream is rigid by design

Chagee says it sources tea and milk directly for domestic operations, with distribution run from HQ. Reported own Yunnan plantations and a narrow menu (tea lattes about 91% of China sales, secondary source) keep sourcing fixed.



New-store forecasts lean on local judgement

Our reading: a new market's opening forecast is judgement-led by a local General Manager, while service is moving toward semi-automation (QR-scan dispensing, Chagee Express).



Why not a marketplace lens

Chagee runs one closed channel, app to store: no third-party sellers, no sell-vs-FBA choice and no online-versus-offline conflict. We leave the marketplace-design topics aside; the franchise GMV revenue share is the one overlap, and it is out of scope.

● INTERACTIVE EXPLAINER

WHERE DOES UNCERTAINTY LAND?

Our reading of public sources, not a Chagee statement. With sourcing fixed, a demand miss can only be absorbed in the store: by the stock it opened with, or by how fast the counter serves. Local sourcing would ease stock risk, but the counter stays exposed either way.

Rigid sourcing

Localised sourcing

Rigid sourcing Chagee today, our reading

Sourcing ● ● ● Fixed: single origin

Distribution ● ● ● Fixed: run from HQ

Opening stock ● ● ● Absorbs the miss

Counter speed ● ● ● Absorbs the miss

Localised sourcing For contrast only

Sourcing ● ● ● Flexes with local suppliers

Distribution ● ● ● Shorter, local

Opening stock ● ● ● Less exposed

Counter speed ● ● ● Still exposed

● ● ● low ● ● ● high exposure to a demand miss

Printed with both columns shown. Live: the toggle highlights one.

OUR VIEW

Take the rigid upstream as given; fix the store-level decisions.

New-market uncertainty lands on three decisions we can model: the demand forecast (slide 2), the opening stock (slide 3) and the counter setup (slide 4).

App-only ordering lets Chagee read a new market's demand before the store opens, and tell opening hype from repeat habit



Every order is tied to a member

With app-only ordering (our understanding), each cup is linked to a member account, so a pre-launch waitlist is a demand signal, not a guess.



More signal, a narrower band

The forecast is a range, not a number. As sign-ups build before launch, the band narrows from $\pm 50\%$ to $\pm 15\%$ of the base (illustrative).



Hype versus habit

Opening crowds are not steady demand: the Kuala Lumpur store reportedly sold 1,000+ cups on its first day (secondary source). Member IDs separate first orders from repeat orders.



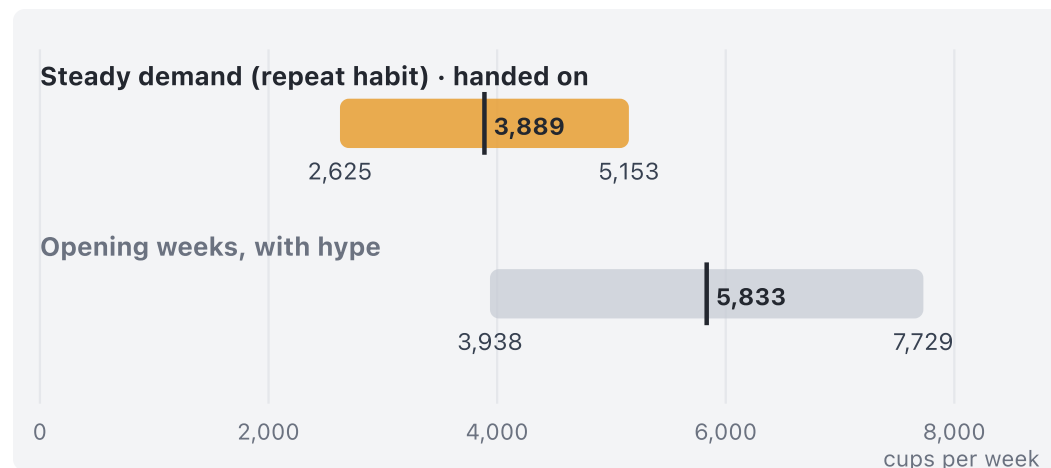
Blind spot: tourists and non-app orders

Visitors and tourist exceptions are outside the member signal, so the forecast understates demand where tourists matter. We name this gap rather than model it.

● INTERACTIVE EXPLAINER

DEMAND RANGE BEFORE OPENING

Weekly cups at one new store. The band covers 80% of likely outcomes; its half-width is the uncertainty that slides 3 to 5 use.



BAND HANDED ON

$\pm 32.5\%$ of base

Low	2,625
Base	3,889
High	5,153
Width	2,528

Cups per week. Opening hype removed: steady demand.

Not in the signal: tourists and non-app orders, so true demand is likely higher.

Pre-launch signal 50% → $\pm 32.5\%$

Opening hype removed

Hype included

Printed at the default signal strength (50%) with opening hype removed.

OUR VIEW

Hand on a demand band, not a single number.

Our contribution, to our knowledge based on public sources: using app-only member data to separate opening hype from repeat demand before launch. Tourist demand stays a named blind spot.

The wider the demand range, the costlier the opening-stock decision, especially with fresh milk



Two ways to be wrong

Too much fresh milk expires; too little turns customers away. Week 1 newsvendor logic: stock up to the point where the next litre is as likely to be wasted as to save a sale.



Here a stockout costs more than waste

A missed cup loses S\$3.82; a spare litre of milk wastes S\$2.55. So the best order sits high in the band, at the 91% point of demand.

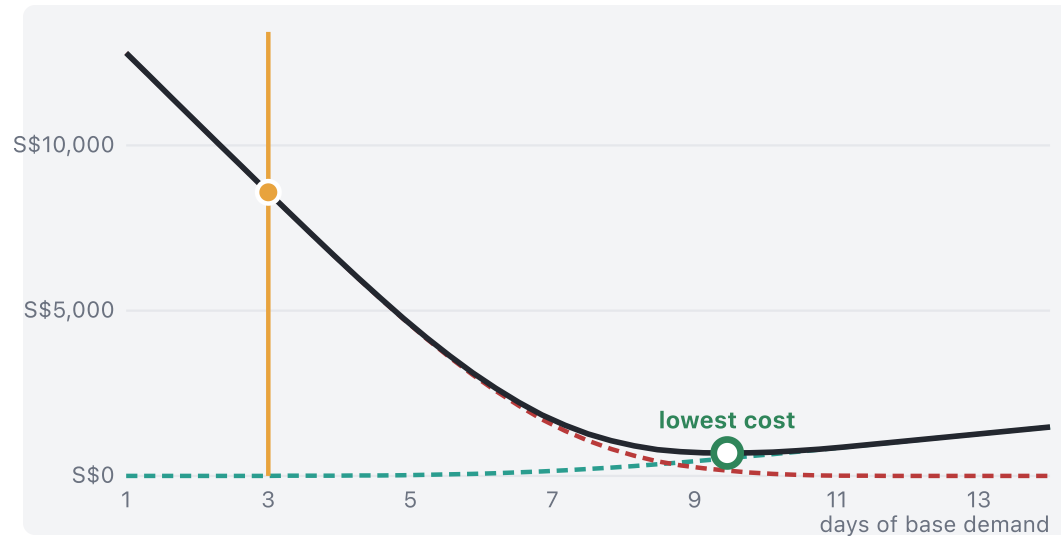


Shelf life caps the buffer

Fresh milk keeps about 7 days (illustrative), so extra stock cannot be carried forward. A narrower band from slide 2 is the cheaper fix.

● INTERACTIVE EXPLAINER OPENING STOCK CALCULATOR

Expected cost of the first fresh-milk order at one store, using slide 2's band. Overseas milk sourcing is unknown; figures are generic.



■ Total expected cost ■ Milk wasted ■ Cups missed

Days of opening stock 3 days

Printed at the band from slide 2 and the default of 3 days of stock.

AT 3 DAYS OF STOCK

S\$8,581 expected

Milk wasted **S\$3**

Cups missed **S\$8,578**

BEST ORDER AT THIS BAND (±32.5%)

9.5 days **S\$694**

Same order, strong signal **S\$320**

Hand-off: open with 788 L of milk. Counter (slide 4): cheapest is All-human; mixed beats full automation by S\$0.25/cup.

OUR VIEW

Narrow the band before you stock the store.

From no signal to the default signal, the expected cost of the best opening order falls from S\$1,064 to S\$694 per store (illustrative). Hand-off to slide 4: the same band sizes the counter.

At a new store's volume, a mixed people-and-machine counter beats full automation, because machines sit idle, not because demand is uncertain

We tested the hypothesis that mixed wins only until demand becomes predictable. Our simulator rejected the uncertainty part.



Machines come in whole units, so they sit idle

Even one machine per station serves about 240 cups an hour against a forecast peak of about 61. The fully automated line is busy only 21% to 22% of open hours.



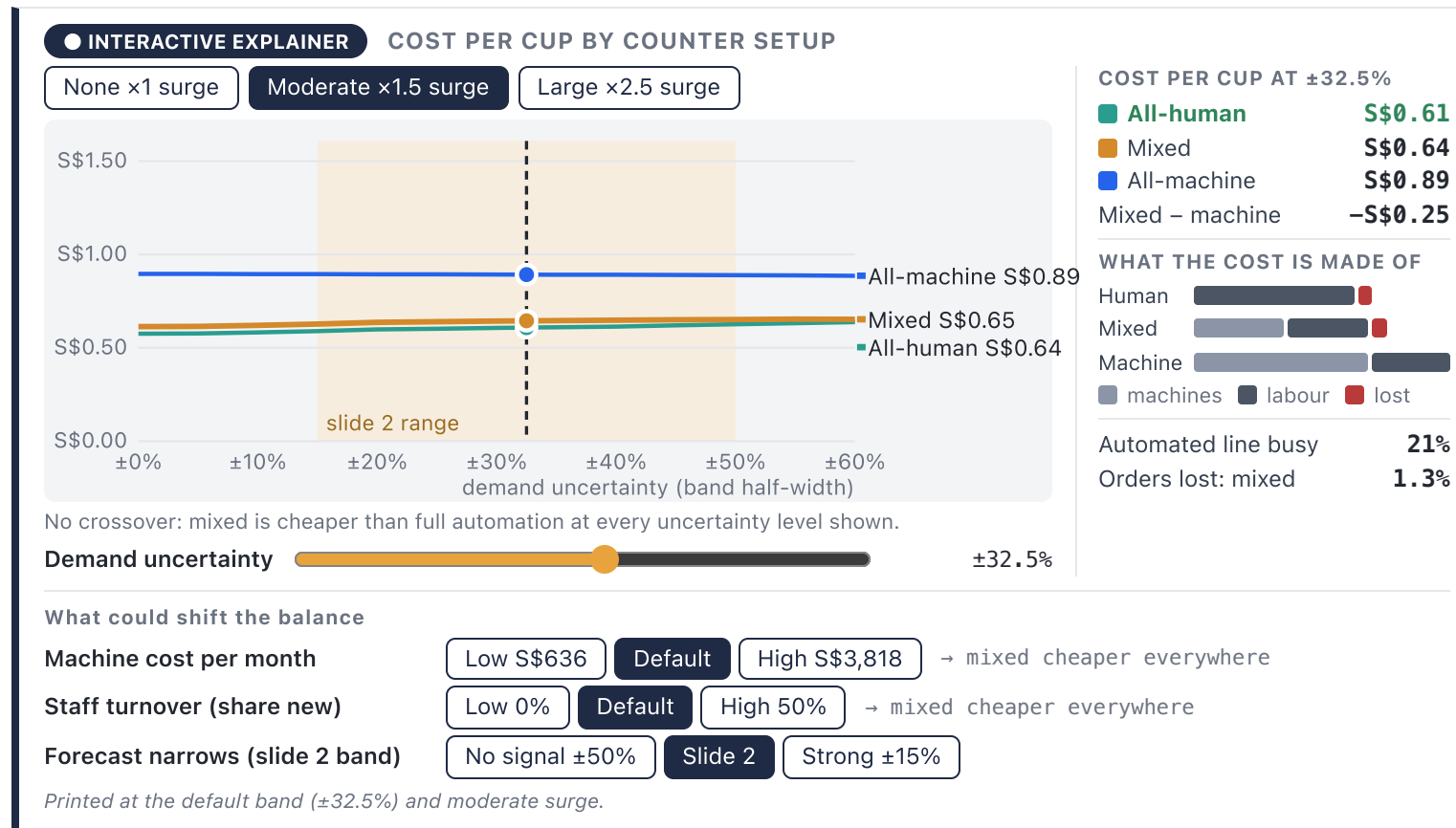
People flex; machines cannot be returned

Rosters follow the hour and the real demand, and staff rebalance across stations like a bucket brigade. A mixed counter buys 3 machines instead of 6, with similar staff hours.



Where machines still win

Big surges: at a 2.5x peak, people-based counters lose about 11% of orders and full automation is cheaper. Also steady high volume, consistency and labour scarcity.



OUR VIEW

Start lean and mixed; automate once volume and peaks are proven, not just once the forecast narrows.

In our model an all-human counter is cheaper still (\$0.61 vs \$0.64 per cup at the default band), but it is the most exposed to surges and turnover. Machine breakdowns are not modelled, which flatters automation.

Even modest per-store savings compound across about 200 overseas openings a year, if the risks are managed

● INTERACTIVE EXPLAINER BENEFIT CALCULATOR

A sizing of the prize, not a forecast: first-year, new stores only, each term an assumption you can move.



Linked: band $\pm 32.5\%$ from slide 2, moderate surge.

Re-link to slides 2–4

≈ S\$10.0m

a year, illustrative

200 openings $\times$ (**200,000** cups $\times$ **S\$0.25** saved per cup + **S\$370** stock cost avoided)

≈ S\$9.9m from the counter + S\$0.1m from opening stock. Saving per cup = full-automation minus mixed cost per cup; stock cost avoided = best-order cost with no signal minus with the slide 2 band.

Printed at the linked defaults: saving per cup from slide 4, waste avoided from slide 3.

OUR CONCLUSION

The saving per cup is small; the case is the count of openings.

Our simulator rejected the claim we set out to prove; what survives is simpler: start lean and mixed, read demand before opening, and automate on proven volume. To our knowledge, based on public sources, simulating a store before it exists is the new part.

RISKS

1 Staff dependence and turnover

A people-based counter relies on trained staff, and new hires are slower. Keep SOP-led training and the flexible worker role; track turnover as in slide 4.

2 Tourist and non-app demand is invisible

The member signal misses visitors, so the band sits too low in tourist areas. Widen it there and log walk-up exceptions after opening.

3 Opening crowds read as steady demand

Sizing to opening-week queues overbuys machines and staff. Separate first orders from repeat orders before committing capex.

4 Automation versus the premium teahouse

Heavy automation could undercut the premium experience. The mixed setup keeps people at the customer-facing steps: finishing and handover.

Next step with real Chagee data

- Replace every assumption with store actuals: station times, wages, machine costs.
- Back-test the waitlist signal on recent openings, such as Korea (launch planned for Q2 2026).
- Pilot mixed versus automated counters in paired new stores.